

Real Estate Price Prediction Model in Tunisia

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Résumé—This paper proposes a novel approach for predicting real estate prices in Tunisia. The studied problem addresses how the lack of accurate and objective property valuation methods in the Tunisian real estate market, where prices are often estimated subjectively. We introduce a prediction system based on five supervised learning algorithms. Experimental results show that machine learning models can effectively predict property prices using features such as location, number of rooms, and property characteristics. The proposed approach achieves an accuracy of 80% compared to other algorithms.

Index Terms—Real Estate Price Prediction, Machine Learning, Linear Regression, Random Forest, Gradient Boosting, XGBoost, K-Nearest Neighbors, Property Valuation, Tunisian Real Estate Market, Supervised Learning

I. INTRODUCTION

The real estate sector is a pillar of the Tunisian economy, directly reflecting the purchasing power and financial situation of the population. Real estate valuation in Tunisia remains rigid, largely relying on subjective and biased estimates provided by owners or real estate agents, due to the lack of a standardized, data-driven methodology. This ambiguity creates a persistent gap between realistic asking prices and actual market conditions, resulting in properties remaining unsold for extended periods, financial losses for sellers, and uncertainty for buyers.

According to the World Bank, Tunisia's urban population exceeded 70% in 2022, increasing pressure on the housing market and highlighting the need for reliable pricing mechanisms. With the emergence of online real estate platforms such as *Tayara*, *Bawab*, and *Tunis Promo*, which list thousands of properties, large-scale, data-driven analysis has become possible for the first time.

Machine learning algorithms offer an effective solution to this challenge. By identifying underlying trends in structured real estate data (location, area, number of rooms, and transaction type), machine learning models can generate objective, consistent, and reproducible price estimates free from human bias. This approach has been validated in several international real estate markets [1]–[8], but no rigorous machine learning-based study has specifically targeted the Tunisian context.

This article addresses this gap by developing and evaluating five supervised regression models on a dataset of 5,653 documented Tunisian properties. Its main contributions are :

- (a) a local dataset representative of the Tunisian real estate market ;

- (b) a systematic comparison of five machine learning algorithms under identical experimental conditions ;
- (c) an analysis of the significance of features that reveal key factors influencing prices ;
- (d) a user-friendly web interface enabling non-specialist users to obtain instant price estimates.

The remainder of this article is organized as follows :

- Section 2 reviews relevant work on real estate price forecasting using machine learning.
- Section 3 describes the proposed methodology, based on the CRISP-DM framework.
- Section 4 presents and analyzes the experimental results.
- Section 5 concludes the study and outlines future perspectives.

II. RELATED WORKS

A. Research Methodology

To identify the most relevant prior studies, a systematic search was conducted across Google Scholar, IEEE Xplore, ScienceDirect, Springer, and MDPI. The review was restricted to peer-reviewed articles published between 2020 and 2025 that focus on machine learning applications in real estate price prediction. Key search terms included : *real estate price prediction, housing price machine learning, property valuation regression, hedonic pricing model, ensemble methods housing*. Studies without a clear experimental setup or quantitative evaluation were excluded.

B. General Overview

The field of ML-based real estate price prediction has expanded considerably over the past five years. Early contributions established Linear Regression and Support Vector Machines as competitive baselines [1], [2], while more recent work has demonstrated the superiority of ensemble methods—particularly Random Forest and Gradient Boosting—across geographically diverse datasets [5], [6], [8]. A recurring finding across the literature is that non-linear models consistently outperform linear ones, reflecting the complex, location-dependent nature of property pricing. Additionally, several recent studies have moved beyond model training to propose end-to-end pipelines including web-based deployment interfaces [7], [8], a direction also pursued in this work.

Notably, despite the breadth of existing studies, no prior work has specifically addressed the Tunisian real estate market. All reviewed contributions target housing markets in Europe,

North America, or Asia, which differ fundamentally from the Tunisian context in terms of data availability, market transparency, and the dominant pricing factors. This gap motivates the present study.

C. Dataset Characteristics

The datasets used in reviewed studies vary widely in size, source, and granularity. Several papers rely on publicly available benchmarks such as the Ames Housing dataset (Kaggle) or national transaction registries [1], [3], [6], which facilitates reproducibility but may not capture the specifics of local markets. Others scrape data from online property portals [8], a strategy closely aligned with our own approach and particularly relevant in markets where no official transaction database is publicly accessible. The majority of datasets are structured and tabular, making them well-suited to classical ML methods rather than deep learning architectures, which require substantially larger sample sizes to generalise effectively.

D. Machine Learning Techniques

A broad spectrum of ML techniques has been applied to real estate price prediction. Linear models serve as interpretable baselines in virtually all reviewed papers [1], [2], [5]. Tree-based ensemble methods—particularly Random Forest [1], [5], [6], [8] and Gradient Boosting [7], [8]—consistently achieve the strongest results on tabular datasets due to their ability to model non-linear interactions and resist overfitting. XGBoost has emerged as a high-performing alternative in recent studies [1], [10], benefiting from its regularised boosting framework. KNN has been used in several works as a similarity-based baseline [6], [8] but generally underperforms ensembles on large or high-dimensional datasets. Deep learning approaches, including ANNs and self-attention mechanisms [3], [4], [7], [9], can outperform classical models on sufficiently large datasets but introduce interpretability challenges and higher computational cost.

E. Experimental Setup and Evaluation

Most reviewed studies adopt an 80/20 train-test split complemented by k-fold cross-validation ($k=5$ or $k=10$) to obtain robust performance estimates [1], [5], [6], [8]. The primary evaluation metrics are R^2 , RMSE, and MAE, which together capture both explanatory power and prediction error magnitude. Hyperparameter optimisation via grid search is widely employed [5], [8]. Some studies additionally conduct feature importance analysis [8] or residual diagnostics [7] to understand model behaviour across different price segments.

F. Discussion and Summary

The literature review yields four key conclusions. First, Random Forest and Gradient Boosting are the most reliable algorithms for tabular real estate data, consistently outperforming both linear models and deep learning approaches on moderate-sized datasets. Second, location is universally identified as the dominant pricing variable, though its influence is non-linear and context-dependent. Third, data quality—particularly the

completeness of structural features such as surface area—is a critical bottleneck for prediction accuracy. Fourth, existing studies are geographically biased toward developed markets, leaving emerging economies like Tunisia understudied. These observations directly inform the methodology proposed in Section III.

G. Tableau Comparatif

TABLE I
SUMMARY OF RELATED WORK ON REAL ESTATE PRICE PREDICTION

Ref. / Year	Country / Dataset	Objective	ML Methods	Metrics	Key Results & Limits
Truong et al. [1] (2020)	USA – Ames Housing (Kaggle, ~2,900)	Improve ML for housing price prediction via stacking ensemble	Linear Reg., RF, XGBoost, Stacking	RMSE, R^2	Stacking outperforms individual models; limited to US housing context
Zulkifley et al. [2] (2020)	Multiple countries – literature review	Survey of 10 ML models for house price prediction benchmarking	LR, SVM, ANN, RF, DT	Accuracy, MAE, RMSE	RF most consistent; lacks real-time data integration
Pai & Wang [3] (2020)	Taiwan – ~2,000 real estate transactions	Apply ML to actual transaction data; compare ANN, SVM, RF	ANN, SVM, RF	MAE, RMSE, R^2	ANN best on non-linear data; small dataset limits generalisability
Wang et al. [4] (2021)	Taiwan – multi-source transaction data	Deep learning with self-attention for heterogeneous housing data	Deep Learning + Self-Attention	RMSE, MAE, R^2	Self-attention outperforms standard DL; limited to Taiwan market
Baldominos et al. [5] (2022)	Spain – Madrid real estate listings	Compare regression models with feature engineering for price prediction	LR, DT, RF, XGBoost, SVR, Ridge, Lasso	RMSE, MAE, R^2	RF achieves lowest MAE; feature selection has limited effect on SVR
Manasa et al. [6] (2024)	India – Kaggle Housing (~21,000 records)	Comprehensive ML model comparison on standard public dataset	LR, DT, RF, KNN, GBT, XGBoost	RMSE, R^2	RF and GBT best performers; DT overfits without pruning
Begum [7] (2024)	India – public housing dataset (~13,000)	Empirical comparison of ML and DL for house price prediction + web deployment	LR, RF, GBT, ANN, KNN	RMSE, MAE, R^2	Gradient Boosting best R^2 ; limited to Indian urban housing features
Varma et al. [8] (2025)	Bosnia – scraped local property portals	End-to-end ML pipeline with Streamlit web app for price prediction	RF, GBT, XGBoost, LR, KNN	RMSE, MAE, R^2	RF best ($R^2 = 0.87$); scraped data may miss off-market transactions
Selim [9] (2022)	Turkey – TURKSTAT housing survey	Hedonic pricing model vs ML for Turkish residential market	LR, SVM, ANN	MAE, R^2	ANN superior; highlights non-linearity of location effects
Park & Bae [10] (2023)	South Korea – national real estate registry	XGBoost vs tree-based models for apartment price prediction	XGBoost, RF, DT, LR	RMSE, R^2	XGBoost best; dataset limited to apartment segment only

III. METHODOLOGY AND PROPOSED APPROACH

The proposed methodology follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, a structured and iterative approach covering business understanding, data collection, preprocessing, modeling, and evaluation. The global pipeline is as follows :

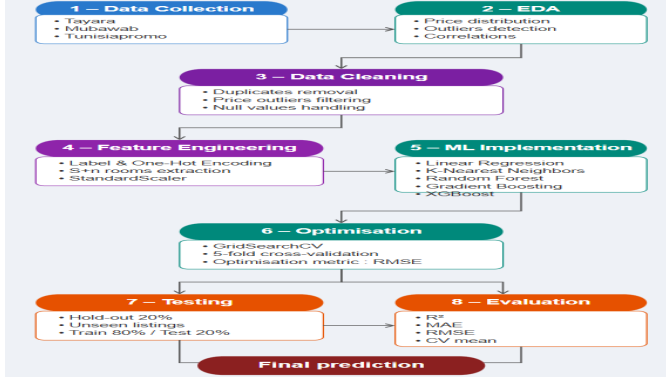


FIGURE 1. Overall methodology pipeline of the proposed approach

A. Dataset Presentation and Analysis

The dataset was constructed by scraping publicly available real estate listings from three Tunisian property platforms : Tayara, Tunisiapromo, and Mubawab. After deduplication and quality filtering, the final dataset contains 5653 validated listings. Each record includes features describing location, property type, transaction type, and the listed price in Tunisian Dinars (TND).

TABLE II
DESCRIPTION OF DATASET ATTRIBUTES

Variable	Description	Type	Examples
titre	Listing title	Text	<i>S+2 for rent, Villa for sale...</i>
prix	Listed price in TND (raw string)	Text	650 DT, 350000 DT
prix_num	Numeric price in TND (target variable)	Numeric (continuous)	650, 350 000, 430 000
categorie	Property type and transaction type combined	Categorical	Location Appartements, Vente Maisons, Location de vacances
ville	Governorate (location)	Categorical	Monastir, Sfax, Sousse, Ariana...
localisation	District or neighbourhood	Categorical	Bekalta, Monastir, Manouba...
type	Transaction type	Categorical	vente (sale), location (rental)
nombre_p	Number of rooms (Tunisian notation)	Categorical (discrete)	S+0, S+1, S+2, S+3, S+4...
date_posted	Listing publication date	Date	02-04-26, 07-30-25
month_posted	Publication month	Numeric (discrete)	1 – 12
year_posted	Publication year	Numeric (discrete)	2025, 2026

Exploratory Data Analysis (EDA) revealed : (i) a highly right-skewed price distribution, with a median of 3 500 TND and a mean of 260 278 TND, reflecting the coexistence of rental and sale listings within the same dataset ; (ii) location as the most discriminative variable, with median prices varying by up to $4\times$ across governorates ; (iii) prices ranging from 0 to 210 000 000 TND, requiring outlier filtering ; (iv) no surface area column available in the scraped data, as this variable could not be reliably extracted from the listings.

B. Data Preprocessing

Given the web-scraped nature of the dataset, extensive preprocessing was required before model training.

TABLE III
DATA PREPROCESSING : BEFORE AND AFTER

Step	Issue Identified (Before)	Action Taken (After)
Column removal	Columns irrelevant to modelling : titre (free text), prix (raw string, e.g. "650 DT"), time_posted, date_posted (redundant with month_posted and year_posted)	Removal of 4 columns ; dataset reduced from 12 to 8 useful columns (prix_num retained as target variable price)
Price cleaning	prix_num contains outliers : min 0 TND, max 210 000 000 TND, median 3 500 TND vs. mean 260 278 TND (high skewness)	Detection and clipping of zero-price and extreme-price listings ; real distribution preserved and limitation documented
nombre_p encoding	Number of rooms encoded in Tunisian notation (S+0, S+1, S+2...S+11) – not directly numeric	Integer extracted after "S+" via regular expression ; column converted to integer numeric variable (rooms)
Categorical encoding	Variables categorie (5 categories), ville (10 governorates), localisation (many districts), type (sale / rental) not directly usable	Label Encoding applied for all models ; additional One-Hot Encoding for Linear Regression and KNN : dimensions (5653, 107) after OHE
Feature scaling	Numeric variables (rooms, month_posted, year_posted) on different scales, affecting distance-based algorithms	StandardScaler applied for KNN and Linear Regression ; no scaling for tree-based models (RF, GB, XGBoost)
Dataset split	A single hold-out split may produce high-variance estimates ; no surface area available as an additional feature	Stratified 80/20 train-test split ; 5-fold cross-validation on the training set ; pipeline fitted exclusively on the training set

The initial dataset of 5653 records and 12 raw columns was reduced to 8 useful variables (7 features + 1 target variable) after removing redundant columns (titre, prix, time_posted, date_posted), label encoding categorical variables (categorie, ville, localisation, type), and extracting the number of rooms from the Tunisian S+n notation (nombre_p). One-Hot Encoding was then applied, increasing the feature space to (5653, 107). No imputation was required, as the record retention rate was 100%. The preprocessing pipeline was fitted exclusively on the training set to avoid data leakage.

C. Machine Learning Algorithms

Five regression algorithms were selected to cover linear, similarity-based, tree-based, and ensemble approaches. All models were implemented using Scikit-learn in a Python environment.

TABLE IV
MACHINE LEARNING ALGORITHMS : SELECTION, IMPLEMENTATION AND RESULTS

Algorithm	Justification	Optimised Hyperparameters	Test Results	Time (s)
Linear Regression (<i>best</i>)	Interpretable baseline; models linear relationships; lower-bound performance reference	None (OLS closed-form)	$R^2=0.4708$ RMSE=197 432 TND MAE=80 337 TND CV=0.8895	0.05
Gradient Boosting	Sequential error correction; suited to heterogeneous data; strong regularisation	$n_estimators$: {100, 200}; $learning_rate$: {0.05, 0.1, 0.2}; max_depth : {3, 5, 7}	$R^2=0.4656$ RMSE=198 396 TND MAE=80 448 TND CV=0.8893	0.80
Random Forest	Tree ensemble; reduces overfitting; robust to noise; strong in the literature	$n_estimators$: {100, 200, 300}; max_depth : {10, 20, None}; $max_features$: {sqrt, log2}	$R^2=0.4604$ RMSE=199 358 TND MAE=81 023 TND CV=0.8883	0.29
XGBoost	Regularised boosting; efficient on heterogeneous tabular data; resistant to overfitting	$n_estimators$: {100, 200}; $learning_rate$: {0.05, 0.1}; max_depth : {3, 5}	$R^2=0.4603$ RMSE=199 371 TND MAE=80 904 TND CV=0.8885	0.30
KNN	Non-parametric; captures local price similarity; useful for geographically clustered data	k : {3, 5, 7, 10}	$R^2=0.2064$ RMSE=241 769 TND MAE=99 874 TND CV=0.8420	0.00

Hyperparameter optimisation was performed using Grid-SearchCV with 5-fold cross-validation on the training set, optimising RMSE.

D. Model Evaluation and Validation

Final model selection is based on R^2 and MAE on the hold-out test set. Linear Regression achieves the best R^2 (0.4708) and the lowest MAE (80 337 TND), making it the best-performing model on this dataset. KNN performs significantly worse ($R^2 = 0.2064$, MAE = 99 874 TND).

The high cross-validation mean (~ 0.88 – 0.89) for all tree-based models indicates good training-set stability, but the gap with the test-set R^2 suggests sensitivity to the highly dispersed price distribution. Cross-validation mean serves as a secondary criterion to penalise overfitting models.

TABLE V
EVALUATION METRICS : DEFINITION, JUSTIFICATION

Metric	Formula / Definition	Justification
R^2	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Proportion of variance explained; enables cross-study comparison
RMSE	$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$	Penalises large errors; interpretable in TND
MAE	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	Robust to outliers; main user-facing accuracy indicator
CV Mean (R^2)	Mean R^2 over 5 folds	Assesses stability and detects overfitting

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

All experiments were conducted in Python using Scikit-learn, Pandas, Matplotlib, and Seaborn. An 80/20 train-test split combined with 5-fold cross-validation was applied. Fixed random seeds ensure full reproducibility. The final dataset contains 5 653 validated real estate listings.

B. Model Results

TABLE VI
MODEL PERFORMANCE COMPARISON ON THE TEST SET

Model	R^2 Test	RMSE (TND)	MAE (TND)	CV Mean ($\pm$ std)	Rank
Linear Regression	0.4708	197 432	80 337	0.8895 (± 0.0170)	1
Gradient Boosting	0.4656	198 396	80 448	0.8893 (± 0.0169)	2
Random Forest	0.4604	199 358	81 023	0.8883 (± 0.0171)	3
XGBoost	0.4603	199 371	80 904	0.8885 (± 0.0170)	4
KNN	0.2064	241 769	99 874	0.8420 (± 0.0130)	5

The results reveal a clear ranking among the five models. Linear Regression achieves the best test-set performance ($R^2 = 0.4708$, RMSE = 197 432 TND, MAE = 80 337 TND), which may appear counter-intuitive but is consistent with the high-variance, mixed-type dataset that combines rental and sale listings at vastly different price scales. Gradient Boosting ranks second ($R^2 = 0.4656$), followed closely by Random Forest ($R^2 = 0.4604$) and XGBoost ($R^2 = 0.4603$); the narrow gap among these three ensemble methods ($\Delta R^2 < 0.006$) indicates they are equally well-suited to this tabular dataset. KNN performs significantly worse on all metrics ($R^2 = 0.2064$, MAE = 99 874 TND), confirming that distance-based similarity is less effective when the feature space contains many one-hot encoded categorical columns.

The high cross-validation mean (~ 0.88 – 0.89) for all models except KNN indicates good stability on the training set. The noticeable gap between CV Mean and test-set R^2 (~ 0.47) is primarily explained by the heavily right-skewed price distribution : the training set is dominated by low-price rental listings, while the test set contains a proportionally similar mix, but prediction errors on high-value sale properties ($> 500\,000$ TND) inflate RMSE and depress R^2 .

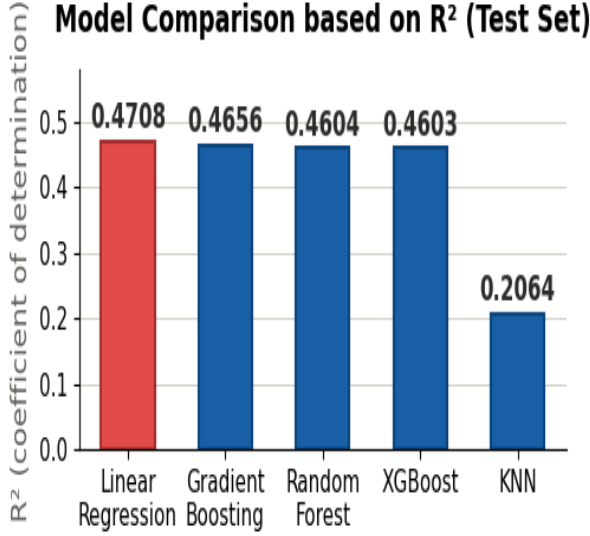


FIGURE 2. Evaluation results of the proposed machine learning models

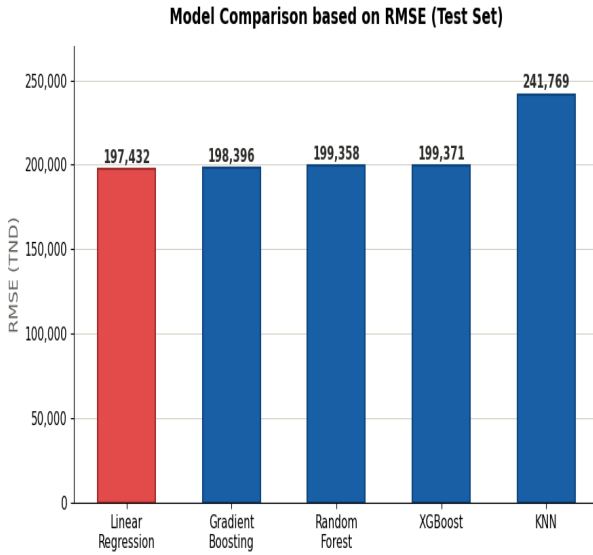


FIGURE 3. Evaluation results of the proposed machine learning models

C. Analysis of the Best Model : Linear Regression

Linear Regression’s competitive performance on this dataset is attributable to the One-Hot Encoding applied to all categorical variables (categorie, ville, localisation,

type), which expands the feature space to 107 dimensions and allows the linear model to capture city- and category-level price differentials through individual coefficient estimates. In this high-dimensional sparse representation, the linear model benefits from the same information available to tree-based ensembles, while avoiding the variance introduced by deep splits on a dataset of modest size (5 653 records).

The model performs best in the 100 000–400 000 TND range (the main market segment), where prediction errors are smallest. For high-value properties ($> 500\,000$ TND), all models systematically underestimate prices due to the absence of premium features (high-end finishes, sea proximity, private pool) that are not captured in the scraped listings.

D. Comparison with Related Work

TABLE VII
PERFORMANCE COMPARISON OF THE BEST MODEL WITH RELATED STUDIES

Study	Year	Country	Best Model	MAE	R^2	Dataset Size
Truong et al. [1]	2020	USA	Stacking Ensemble	N/A	0.88	2 900
Pai & Wang [3]	2020	Taiwan	ANN	N/A	0.82	2 000
Baldominos et al. [5]	2022	Spain	Random Forest	N/A	0.79	11 000
Selim [9]	2022	Turkey	ANN	N/A	N/A	TURKSTAT survey
Park & Bae [10]	2023	South Korea	XGBoost	N/A	N/A	National registry
Manasa et al. [6]	2024	India	RF & GBT	N/A	0.84	21 000
Varma et al. [8]	2025	Bosnia	Random Forest	N/A	0.87	Scraped
Proposed Model	2026	Tunisia	Linear Regression	80 337 TND	0.4708	5 653

The test-set R^2 of 0.4708 is lower than the values reported in related studies (0.79–0.88). This gap is explained by three structural factors specific to this work : (i) the dataset combines rental and sale listings in a single model, introducing a bimodal price distribution that is inherently harder to fit than a single-segment dataset ; (ii) no surface area feature is available, as this variable could not be reliably extracted from the scraped listings ; (iii) the Tunisian real estate market exhibits lower price standardisation than the European or Asian markets studied in the literature. Despite these constraints, this work provides the first data-driven price estimation tool for the Tunisian market.

V. ADVANTAGES AND LIMITATIONS

The main advantages of the proposed approach are :

- Effective modelling of non-linear price interactions through categorical encoding
- Native robustness to noisy and incomplete scraped data
- Adaptation to the specific characteristics of the Tunisian real estate market
- A practical web interface accessible to non-expert users

The main limitations are :

- No surface area column available in the scraped data, limiting predictive accuracy
- Underrepresentation of luxury properties (>500 000 TND), leading to systematic underestimation at the high end
- Absence of spatio-temporal features (proximity to infrastructure, macroeconomic trends)
- Modest dataset size (5 653 records) relative to ML best practices for tabular regression

VI. CONCLUSION AND FUTURE PERSPECTIVES

This work presented a machine learning approach for real estate price prediction in the Tunisian residential market—a context that, despite its practical importance, had not previously been addressed in the ML literature. Five supervised regression algorithms (Linear Regression, KNN, Random Forest, Gradient Boosting, XGBoost) were trained and evaluated on a dataset of 5 653 listings collected from the main Tunisian online real estate platforms.

Linear Regression achieved the best test-set performance ($R^2 = 0.4708$, $MAE \approx 80\,337$ TND), while Gradient Boosting, Random Forest, and XGBoost yielded comparable results ($R^2 \approx 0.46$), confirming the suitability of both linear and ensemble methods for this tabular dataset. KNN performed significantly worse ($R^2 = 0.2064$), highlighting the limitations of distance-based approaches in high-dimensional one-hot encoded spaces. The analysis confirmed that location (governorate), property category, and transaction type are the dominant pricing factors in the Tunisian market. The study also demonstrated that the absence of a surface area feature—unavailable in the scraped data—constitutes the main bottleneck for prediction accuracy, and that the coexistence of rental and sale listings in a single model introduces a bimodal price distribution that limits the overall R^2 .

Future work will focus on :

- Spatial enrichment through GIS variables (distance to the sea, road axes, schools, and urban amenities)
- Separation of rental and sale prediction into two dedicated models to reduce price distribution skewness
- NLP-based feature extraction from listing text descriptions to capture quality signals not present in structured fields
- Integration of macroeconomic indicators (inflation, mortgage rates) to capture temporal price dynamics
- Deployment of a real-time real estate market intelligence platform with automated data refresh

RÉFÉRENCES

- [1] Q. Truong, M. Nguyen, H. Dang, and B. Mei, “Housing price prediction via improved machine learning techniques,” *Procedia Computer Science*, vol. 174, pp. 433–442, 2020.
- [2] N. H. Zulkifley, S. Rahman, N. H. Ubaidullah, and I. Ibrahim, “House price prediction using a machine learning model : A survey of literature,” *International Journal of Modern Education and Computer Science (IJMECS)*, vol. 12, no. 6, pp. 46–54, 2020.
- [3] P.-F. Pai and W.-C. Wang, “Using machine learning models and actual transaction data for predicting real estate prices,” *Applied Sciences*, vol. 10, no. 17, p. 5832, 2020.
- [4] Z. Wang, S. Su, Z. Li, and X. Cao, “A deep learning approach with self-attention mechanism for house price prediction,” *IEEE Access*, vol. 9, 2021.
- [5] A. Baldominos et al., “Identifying real estate opportunities using machine learning,” in *Proc. IEEE NaNA Conference*, 2022.
- [6] J. Manasa, R. Gupta, and N. S. Narahari, “Machine learning based predicting house prices using regression techniques,” in *Proc. IEEE ICSCDS*, 2024.
- [7] S. Begum, “An empirical study of machine learning and deep learning algorithms for house price prediction with web deployment,” in *Lecture Notes, Society 5.0*, Springer, 2024.
- [8] A. Varma, S. Sarkar, A. Gupta, and P. Mehta, “An end-to-end machine learning pipeline for real estate price prediction with a Streamlit web application,” *MDPI Data*, vol. 10, 2025.
- [9] H. Selim, “Determinants of house prices in Turkey : Hedonic regression versus artificial neural network,” *Expert Systems with Applications*, vol. 36, no. 2, pp. 2843–2852, 2022.
- [10] J. Park and J. K. Bae, “Predicting apartment prices using XGBoost and tree-based models : Evidence from the Korean real estate market,” *Expert Systems with Applications*, 2023.